

Performance of phase transform for detecting sound sources with microphone arrays in reverberant and noisy environments

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Abstract

The performance of sound source location (SSL) algorithms with microphone arrays can be enhanced by processing signals prior to the delay and sum operation. The phase transform (PHAT) has been shown to improve SSL images, especially in reverberant environments. This paper introduces a modification, referred to as the PHAT- β transform, that varies the degree of spectral magnitude information used by the transform through a single parameter. Performance results are computed using a Monte Carlo simulation of an eight element perimeter array with a receiver operating characteristic (ROC) analysis for detecting single and multiple sound sources. In addition, a Fisher's criterion performance measure is also computed for target and noise peak separability and compared to the ROC results. Results show that the standard PHAT significantly improves detection performance for broadband signals especially in high levels of reverberation noise, and to a lesser degree for noise from other coherent sources. For narrowband targets the PHAT typically results in significant performance degradation; however, the PHAT- β can achieve performance improvements for both narrowband and broadband signals. Finally, the performance for real speech signal samples is examined and shown to exhibit properties similar to both the simulated broad and narrowband cases, suggesting the use of β values between 0.5 and 0.7 for array applications with general signals.

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1. Introduction

Sound source location (SSL) with microphone arrays has important applications in multi-modal tracking and advanced human–computer interaction (HCI) [1–5]. While several approaches exist for SSL, the Steered Response Power (SRP) algorithm

typically exhibits the most robust performance [6,7]. The SRP algorithm creates a likelihood function [8] over a field of view (FOV), through which SSLs can be identified by local maxima. Previous works have measured performance in terms of location error, target-to-noise amplitude enhancements, and percent anomalous detections [6–8]; however, in many practical situations the sound sources enter and leave a FOV over time, and switch between active and inactive states. Therefore, before processing for the purpose of tracking or location estimation, the existence of the sound source must first be detected.

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This paper formulates the SRP problem in terms of detection rather than location estimation. Detection performance is assessed using the area under the receiver operating characteristics (ROC) curve [9–11], and Fisher's criterion (FC) [12]. The ROC analysis, originally proposed for assessing radar systems [11], indicates detection system performance independent of a specific threshold. The ROC curve plots true-target versus false-target detection probabilities by effectively applying all thresholds over the range of SRP values from the two conditions. The area under the ROC is useful for comparing detection systems and has been well studied, especially for applications in medical imaging [9,10]. The FC is the ratio of the squared difference of the means of the target and noise peaks divided by the sum of their variances, and represents a figure of merit more directly related to target-to-noise amplitude ratios. The FC is computed and used in this paper to illustrate the differences between directly assessing detection (with the ROC areas) as opposed to enhancing the distance between the mean target and noise peaks in the SRP likelihood function, which more directly correspond to SNR enhancement.

The heart of the SRP is the filter-and-sum (or delay-and-sum) beamforming operation, which reduces noise power proportional to the number of microphone channels. Uncorrelated noise typically results from the independent (electronic) noise on each microphone channel. Correlated noise, on the other hand, results from coherent noise sources in the room, such as other sound sources and reverberations. Correlated noise presents greater challenges for SSL problems, and therefore will be the primary noise considered. Approaches to deal with correlated noise have included various types of spectral weighing involving the generalized cross correlation (GCC). If the noise spectrum is known, maximum likelihood weights can be developed to deemphasize low SNR spectral regions [7,13]. If the noise spectrum is not known, a phase transform (PHAT) can be applied that effectively whitens the signal spectrum [6,7,13–16].

While previously published results demonstrate that the PHAT processing generates superior acoustic images with sharper targets and an attenuated noise field [6–8], there have been no studies to see how well targets of interest can be separated from noise peaks through an automatic threshold procedure. In addition, there has been no detailed evaluation of relationship between the

nature of the signal bandwidth and the PHAT performance. This paper introduces a parametric modification of the PHAT to study the influence of the magnitude spectra on performance. Results from Monte Carlo simulations are presented that compare implementations of the parametric PHAT implementation for single and multiple impulse-like targets, ranging from narrow to wideband, as well as sample speech signals.

In Section 2 the SRP algorithm is presented for targets in noisy and reverberant environments. The PHAT- β is explicitly defined. Section 3 derives expressions for the expected values and variances of target and noise magnitudes, and introduces the performance measures. Section 4 describes the simulation design along with SRP image examples. Section 5 presents and discusses the results, and Section 6 summarizes the findings and conclusions.

2. SRP algorithm

This section formulates the SRP-PHAT algorithm and identifies parameters affecting performance. The signal and array models are directly used in the simulations.

Consider a set of microphones and sound sources at different spatial locations. Let $u_i(t; \vec{r}_i)$ be the pressure wave resulting from the i th source located at position $\vec{r}_i$. The waveform received by the p th microphone is given by

$$\begin{aligned} v_{p,i}(t; \vec{r}_p, \vec{r}_i) &= \int_{-\infty}^{\infty} h_{p,i}(\lambda; \vec{r}_p, \vec{r}_i) u_i(t - \lambda; \vec{r}_i) d\lambda \\ &+ \sum_{k=1}^K \int_{-\infty}^{\infty} h_{p,k}(\lambda; \vec{r}_p, \vec{r}_k) n_k(t - \lambda; \vec{r}_k) d\lambda + n_p(t), \end{aligned} \quad (1)$$

where $h_{p,i}(\cdot)$ represents the impulse response of the microphone and propagation path from $\vec{r}_i$ to $\vec{r}_p$, $n_k(t)$ represents correlated noise sources located at $\vec{r}_k$, and $n_p(t)$ represents uncorrelated electronic noise.

The coherent noise sources $n_k(t)$ arise from sources not at the point of interest as illustrated in Fig. 1. The broken-line rectangle denotes the FOV containing two targets of interest (i.e. two speakers talking simultaneously). Note that while estimating the SRP at one target location, the other target within the FOV acts as an interfering noise signal and vice versa. The solid rectangle denotes the room

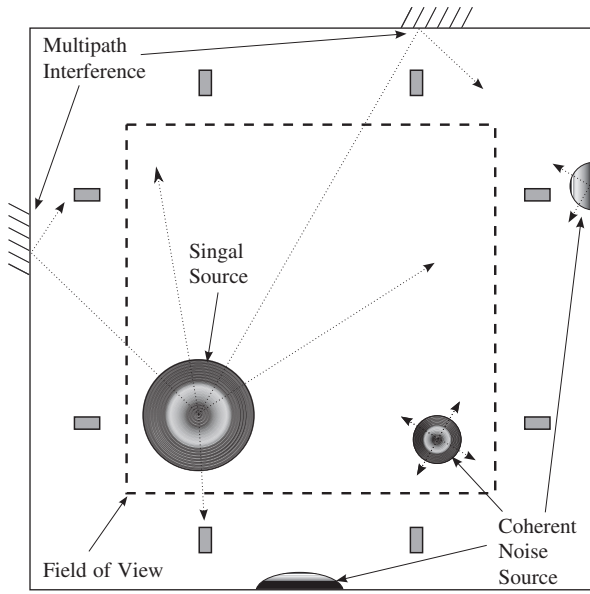


Fig. 1. Target at point of interest with interfering sources in a perimeter microphone array system.

walls that generate reverberations, and transmit independent noise sources, such as traffic noise through a window or turbulence in air vents. The small gray rectangles denote positions of the microphone array elements.

For reverberant rooms the impulse response can be separated into a signal (direct path) and noise component (reflected path). The impulse response can be expanded as

$$h_{p,i}(t; \vec{r}_p, \vec{r}_i) = h_{p,i}(t) = \sum_{n=0}^{\infty} a_{p,i,n}(t - \tau_{p,i,n}), \quad (2)$$

where $a_{p,i,n}(t)$ denotes the effective impulse response for the n th path between the source at $\vec{r}_i$ and microphone at $\vec{r}_p$, with corresponding path delay $\tau_{p,i,n}$. The path corresponding to $n = 0$ is the direct path between the source and microphone. Let the time interval associated with the l th SRP frame be denoted as Δ_l . The SRP estimate from signal segments in Δ_l can be expressed in the frequency domain (substitute Eq. (2) into (1)) as

$$\begin{aligned} \hat{V}_{p,l}(\omega) = & \sum_{i=1}^{N_T} \hat{U}_{i,l}(\omega) \sum_{m|\tau_{p,i,m} \in \Delta_l} \hat{A}_{p,i,m}(\omega) \exp(j\omega\tau_{p,i,m}) \\ & + \sum_{k=1}^K \hat{N}_k(\omega) \sum_{m|\tau_{p,k,m} \in \Delta_l} \hat{A}_{p,k,m}(\omega) \\ & \times \exp(j\omega\tau_{p,k,m}) + \hat{N}_p(\omega), \end{aligned} \quad (3)$$

where $\hat{U}_{i,l}(\omega)$ is the Fourier transform of the i th source, $u_i(t)$, over the l th interval, while $\hat{N}_k(\omega)$ and $\hat{N}_p(\omega)$ are the Fourier transforms of the correlated and uncorrelated noise sources, respectively. N_T is the number of target sources, K is the number of noise sources, and the inner summation indices, m , denote summing the signal components contributing to the Δ_l interval. For this paper the PHAT is generalized to parametrically control the level of influence from the magnitude spectrum. This transform will be referred to as the PHAT- β and defined as

$$\hat{\theta}_{p,l}(\omega, \beta) = \frac{|\hat{V}_{p,l}(\omega)|}{|\hat{V}_{p,l}(\omega)|^\beta} \angle \hat{V}_{p,l}(\omega), \quad (4)$$

where β varies between 0 and 1. When $\beta = 1$, Eq. (4) becomes the conventional PHAT and the modulus of the Fourier transform becomes 1 for all frequencies, when $\beta = 0$ the PHAT- β has no effect on the original signal.

To develop a signal and noise analysis, the signal term can be separated from the noise terms of Eq. (3) to result in a direct path signal term

$$\hat{v}_{p,i,l}(\omega) = \hat{U}_{i,l}(\omega) \hat{A}_{p,i,0}(\omega) \exp(j\omega\tau_{p,i,0}), \quad (5)$$

and the remaining multi-path signal and independent noise terms

$$\begin{aligned} \hat{h}_{p,i,l}(\omega) = & \hat{U}_{i,l}(\omega) \sum_{\substack{m|\tau_{p,i,m} \in \Delta_l \\ m \neq 0}} \hat{A}_{p,i,m}(\omega) \exp(j\omega\tau_{p,i,m}) \\ & + \sum_{k=1}^K \hat{N}_k(\omega) \sum_{k|\tau_{p,k,m} \in \Delta_l} \hat{A}_{p,k,m}(\omega) \\ & \times \exp(j\omega\tau_{p,k,m}) + \hat{N}_p(\omega). \end{aligned} \quad (6)$$

Apply the PHAT- β to Eq. (6) to express as separate signal and noise terms to obtain

$$\begin{aligned} \hat{\theta}_{p,i,l}(\omega, \beta) = & \frac{|\hat{v}_{p,i,l}(\omega)|}{|\hat{V}_{p,l}(\omega)|^\beta} \exp(j(\angle \hat{U}_{i,l}(\omega) + \omega\tau_{p,i,0})) \\ & + \frac{|\hat{h}_{p,i,l}(\omega)|}{|\hat{V}_{p,l}(\omega)|^\beta} \exp(j\angle \hat{h}_{p,i,l}(\omega)). \end{aligned} \quad (7)$$

Note that the PHAT weighting does not guarantee an SNR enhancement, since the weights depends on the relative spectral magnitudes and not on the expected SNR of the unprocessed signal. The simulations in the next section identify cases when the PHAT enhances and degrades sound source detection performance.

The SRP can be expressed as a filter and sum operation corresponding to a pixel at $\vec{r}_i$. The power

from each individual microphone signal is not directly related to source position coherence, and is therefore subtracted out from the filter and sum [6]. This results in the coherent power, and in the discrete frequency domain can be expressed as

$$S_{i,l}(\beta) = \frac{\Delta_\omega}{T} \sum_{k=K_1}^{K_2} \left(\left| \sum_{p=1}^P \hat{B}_{p,i} \hat{\theta}_{p,i,l}(\omega_k, \beta) \right|^2 - \sum_{p=1}^P |\hat{B}_{p,i} \hat{\theta}_{p,i,l}(\omega_k, \beta)|^2 \right), \quad (8)$$

where T is the length of Δ_l , K_1 and K_2 are indices corresponding to the upper and lower frequencies of the signal (or usable bandwidth), Δ_ω is the frequency domain sampling interval, and $\hat{B}_{p,i}$ is the complex weight representing the delay and filtering associated with the image location and array geometry. The coherent power values $S_{i,l}(\beta)$ become samples/pixels of the SRP likelihood function. For the computations of this paper the magnitudes of $\hat{B}_{p,i}$ were set equal to the reciprocal distance between the p th microphone and the i th SRP pixel location, and normalized by the sum of reciprocal distances over all array elements, in accordance with a nearfield design [17,18] (also is referred to as the weighted SRP [7]).

The signal and noise contributions after the delay and sum operation are obtained by substituting Eq. (7) into Eq. (8):

$$\begin{aligned} S_{i,l}(\beta) = & \frac{\Delta_\omega}{T} \sum_{k=K_1}^{K_2} \sum_{p=1}^P \sum_{q=1, q \neq p}^P \\ & \times [\bar{v}_{p,i,l}(\omega_k) \bar{v}_{q,i,l}(\omega_k) \exp(j\omega_k(\varepsilon_{p,i} + \varepsilon_{q,i})) \\ & + \bar{v}_{p,i,l}(\omega_k) \bar{\eta}_{q,i,l}(\omega_k) \exp(j(\omega_k(\varepsilon_{p,i} + \tau_{p,i,0}) \\ & - \angle \hat{\eta}_{q,i}(\omega_k))) + \bar{\eta}_{p,i,l}(\omega_k) \bar{\eta}_{q,i,l}(\omega_k) \\ & \times \exp(j(\angle \hat{\eta}_{p,i}(\omega_k) - \angle \hat{\eta}_{q,i}(\omega_k)))], \quad (9) \end{aligned}$$

where the $\varepsilon_{p,i}$ term represents the residual phase between the microphone signal and the phase of the steering weights $\hat{B}_{p,i}$. Notation of Eq. (9) was reduced by combining the following terms:

$$\bar{v}_{p,i,l} = \frac{|\hat{B}_{p,i} \hat{v}_{p,i,l}(\omega_k)|}{|\hat{V}_p(\omega_k)|^\beta} \quad \text{and} \quad \bar{\eta}_{p,i,l} = \frac{|\hat{B}_{p,i} \hat{\eta}_{p,i,l}(\omega_k)|}{|\hat{V}_p(\omega_k)|^\beta}. \quad (10)$$

3. Performance metrics

This section identifies critical sources of variability for estimating the SRP likelihood function.

These are randomly generated in the Monte Carlo runs. Performance metrics estimated from the simulation are also developed in this section.

The expected SRP amplitude associated with a target position is obtained by applying the expected value operator to Eq. (9) with the assumptions that at each frequency the residual phase for non-target source locations is uniformly distributed over the interval $[-\pi, \pi]$, and that the residual phases for the target (coherent source) are uniform over the interval $[-\delta, \delta]$, where $\delta \ll \pi$. The δ -interval accounts for the quantization error of the SRP grid spacing. In general, the grid sampling must be fine enough such that $\delta \omega_k \ll \pi$ holds over a significant portion of the target signal spectrum, so that independent of where the target is positioned in the FOV, it will be coherently summed on the nearest grid point. The variations in signal amplitude and phases are the critical components affecting performance, and therefore will be modeled as random variables (RV). The expected value operator in the following analysis will be applied with respect to these quantities. The uniform phase assumption over the 2π -interval results in the noise terms and cross noise–target terms going to zero under the expected value operation over all microphone pairs. Therefore, after applying the expected value operator over both signal magnitudes and phases, only the first term of Eq. (9) remains. The expected SRP value at the target pixel becomes

$$\begin{aligned} \mu_{S_{i,l}|H_1} &= E[S_{i,l}(\beta) | |\hat{U}_i| > 0] \\ &= \frac{(P^2 - P)\Delta_\omega}{T} \sum_{k=K_1}^{K_2} \langle \bar{v}_{p,i,k} \bar{v}_{q,i,k} \rangle_{q \neq p} \text{sinc}(\omega_k \delta), \quad (11) \end{aligned}$$

where the angle brackets denote the mean value for all target signal magnitude pair products in the microphone array, and H_1 denotes the condition that a target source corresponds to the pixel.

Note that when the argument of the sinc function, $\omega_k \delta$, approaches π , the expected value of the target amplitude goes to zero (condition becomes the same as the incoherent case). Since this condition depends on the target signal frequency and grid resolution, a relationship can be derived to limit the variability for a worst-case frequency term due to the quantization of the pixel grid. Consider one frequency term of Eq. (11) and substitute in half the maximum distance between neighboring grid

points for δ to obtain

$$\mu_g(\omega_k, \Delta_{\text{grid}}) = \frac{(P^2 - P)\Delta_\omega}{T} \langle \bar{v}_{p,i,k} \bar{v}_{q,i,k} \rangle_{q \neq p} \times \text{sinc} \left(\frac{\omega_k \sqrt{d \Delta_{\text{grid}}^2}}{2} \right), \quad (12)$$

where d is the dimension of the grid space, and Δ_{grid} is the spacing between grid points along a single dimension (assuming equal spacing in all dimensions). The maximum expected value occurs for $\Delta_{\text{grid}} = 0$, therefore the expected loss from the maximum value can be expressed in terms of the quantization loss ratio

$$Q(\Delta_{\text{grid}}) = \frac{\mu_g(\omega_k, \Delta_{\text{grid}})}{\mu_g(\omega_k, 0)} = \text{sinc} \left(\frac{\omega_k \sqrt{d \Delta_{\text{grid}}^2}}{2} \right). \quad (13)$$

To determine the required grid spacing, given a lower bound on the quantization loss ratio, the inverse sinc function resulting from equation (13) must be computed iteratively. Consider a case where a maximum 3dB loss can be tolerated from grid quantization over a 2-D FOV. To ensure this over the frequencies of interest, use the worst-case frequency (highest frequency in the signal) and compute the resulting grid spacing bound to obtain

$$\Delta_{\text{grid}} \leq \frac{0.4422c}{\sqrt{d}f_h}, \quad (14)$$

where c is the speed of sound and f_h is the frequency at which this bound was set. For all frequency components less than this, the expected quantization loss will not exceed 3 dB.

The variance of the target pixel results from both quantization and noise source contributions. With the same assumptions used in computing the expected value of Eq. (11), the variance of the SRP value from Eq. (9) is

$$\begin{aligned} \sigma_{S_{i,l}|H_1}^2 &= \text{var}[S_{i,l}(\beta) | \hat{U}_i] \\ &= \left(\frac{(P^2 - P)\Delta_\omega}{\Delta_l} \right)^2 \sum_{k=K_1}^{K_2} (1 - \text{sinc}^2(\omega_k \delta)) \\ &\quad \times \langle \bar{v}_{p,i,k} \bar{v}_{q,i,k} \rangle_{q \neq p}^2 + \langle \bar{v}_{p,i,k} \bar{\eta}_{q,i,k} \rangle_{q \neq p}^2 \\ &\quad + \langle \bar{\eta}_{p,i,k} \bar{\eta}_{q,i,k} \rangle_{q \neq p}^2. \end{aligned} \quad (15)$$

The first term within the summation accounts for the variance due to the grid quantization, the middle

term results from correlations between signal and noise magnitudes, and the last term results from the noise sources and directly relates to the sum of the off-diagonal magnitudes of the coherence matrix.

For the performance analyses, pixel locations corresponding to a target will be denoted with condition H_1 , and locations corresponding to no target will be denoted with condition H_0 . The mean and variance of the non-target positions can be obtained by letting the target energy go to zero in Eqs. (11) and (15) to obtain a mean value of

$$\mu_{S_{i,l}|H_0} = E[S_{i,l}(\beta) | \hat{U}_i = 0] = 0 \quad (16)$$

and variance

$$\begin{aligned} \sigma_{S_{i,l}|H_0}^2 &= \text{var}[S_{i,l}(\beta) | \hat{U}_i = 0] \\ &= \left(\frac{(P^2 - P)\Delta_\omega}{T} \right)^2 \sum_{k=K_1}^{K_2} \langle \bar{\eta}_{p,i,k} \bar{\eta}_{q,i,k} \rangle_{q \neq p}^2. \end{aligned} \quad (17)$$

In order to quantify the input signal power relationship, the array SNR will be defined as the power ratio of the direct path of the i th target signal over all other sources and reflections received over the microphone array, given by

$$G_i = \frac{\sum_{p=1}^P \sum_{k=K_1}^{K_2} |\hat{v}_{p,i,k}|^2}{\sum_{p=1}^P \sum_{k=K_1}^{K_2} |\hat{\eta}_{p,i,k}|^2}. \quad (18)$$

The FC for SRP likelihood functions with H_0 and H_1 pixels is defined as

$$f_S(\beta) = \frac{(\mu_{S|H_1} - \mu_{S|H_0})^2}{\sigma_{S|H_1}^2 + \sigma_{S|H_0}^2}. \quad (19)$$

For a discrete set of SRP pixel values ($S_{i,l}$ from Eq. (8)) from both H_1 and H_0 , the ROC curve area can be computed from the function defined as

$$C(S_{k,l|H_0}, S_{i,l|H_1}) = \begin{cases} 1 & \text{for } S_{k,l|H_0} < S_{i,l|H_1}, \\ 0.5 & \text{for } S_{k,l|H_0} = S_{i,l|H_1}, \\ 0 & \text{for } S_{k,l|H_0} > S_{i,l|H_1}, \end{cases} \quad (20)$$

where i and k represent independent estimates of the SRP image pixels under the indicated hypotheses. Given n_1 pixels from H_1 , and n_0 pixels from H_0 , the ROC area can be estimated directly from the pixel amplitudes using the following Wilcoxon statistic [9]:

$$A_z = \frac{1}{n_1 n_0} \sum_{k=1}^{n_1} \sum_{l=1}^{n_0} C(S_{k|H_0}, S_{i|H_1}) \quad (21)$$

and the standard error for the estimated A_z can be computed from

$$\sigma_{SE} \approx \sqrt{\frac{A_z(1 - A_z) + (n_0 - 1)(Q_0 - A_z^2) + (n_1 - 1)(Q_2 - A_z^2)}{n_0 n_1}}, \quad (22)$$

where

$$Q_1 = \frac{A_z}{2 - A_z} \quad \text{and} \quad Q_2 = \frac{2A_z^2}{1 + A_z}. \quad (23)$$

The formulation of A_z as a Wilcoxon statistic also suggests a meaning for the A_z beyond that of a simple figure of merit. An A_z of 0.9, for example, implies a 0.9 probability for a target value exceeding an independent noise value randomly selected from the SRP image. For the analyses in this paper, performance statistics are computed only from positive peak amplitudes in the SRP image. This increases the likelihood that independent values will be used, since pixels involved in gradients between peaks will not be included.

4. Simulation design

This section describes the design of the simulation used to analyze the impact of the β parameter on the performance measures described in the previous section. Monte Carlo simulations are developed to estimate performance over a range of signal sources in coherent and reverberant noise. The simulation models include room effects and targets of different bandwidths to represent extreme cases of natural signals.

Room properties were selected based on an actual laboratory system. The details are summarized in Table 1. A maximum reflection coefficient of 0.95 was assigned to the walls, and 0.9 to the ceiling and floor. In the simulations these values were scaled from 0% to 95% to simulate different levels of room reverberations. The room impulse response

for a given target position was computed using the image method described in [19]. In addition, the frequency-dependent attenuation due to air was accounted for by zero-phase filtering all paths from the source to microphone (including paths from the image scatterers). For each signal path the slope of the log-linear filter spectrum was obtained using the air absorption factor. The impulse response for multi-path component was obtained from this spectrum and used to compute the FIR filter coefficients (order 50 for 16 kHz sampling) using a windowing (Hamming) method [20]. For all simulations an eight microphone perimeter array was used, as shown in Fig. 1. Each microphone was omnidirectional and was placed in the room 25 cm perpendicular from the wall surface.

For each Monte Carlo run the target sources were placed with a uniform distribution in the FOV with a square dimension of 2.4 m centered within the image plane. Two coherent white noise sources were independently generated and placed with a uniform distribution within the room and outside a square boundary 15 cm from the wall. This positioning ensured that a noise source would be outside the FOV and not fall directly on a microphone. Two target signal types were created as impulse responses to a Butterworth filter with a lower 3 dB cutoff of 400 Hz in both cases, and upper cutoff frequency of 600 Hz for the narrowband signal, and 5600 Hz for broadband signal. The Butterworth impulse response was chosen because it has a maximally flat spectrum, while its impulse response is a causal signal with the appropriate phase spectrum. This signal generation resulted in an impulse-like signal from which narrow and broadband signal performance could be inferred. In addition, a real signal was recorded from an adult female saying the word “shoe.” Experiments with the real signal provided a comparison with Butterworth signals, in that the fricative, SH, has a broadband spectrum, and the voiced part, UW, has a narrowband spectrum. Signals were sampled at 16 kHz. The grid spacing was set at 0.02 m, which according to Eq. (14) implies that the quantization losses for frequency components less than 5.4 kHz will not exceed 3 dB.

Fig. 2 shows imaging examples of a single simulation run for a FOV containing two narrow and broadband targets, respectively, for three different β values. These simulations included two coherent noise sources with reverberations from walls having a reflection coefficient of 0.9 and the floor and ceiling having a reflection coefficient of

Table 1
Room and array simulation parameters

Room property	Parameters
Length and width	3.62 m
Height	2.22 m
Mean surface reflection coefficients	3 cases [0, 0.75, 0.89]
Air absorption factor	−32.8 μdB/(m Hz)
Sound speed	348 m/s
Microphone geometry	Equilateral hexagon
Microphone spacing	1.29 m

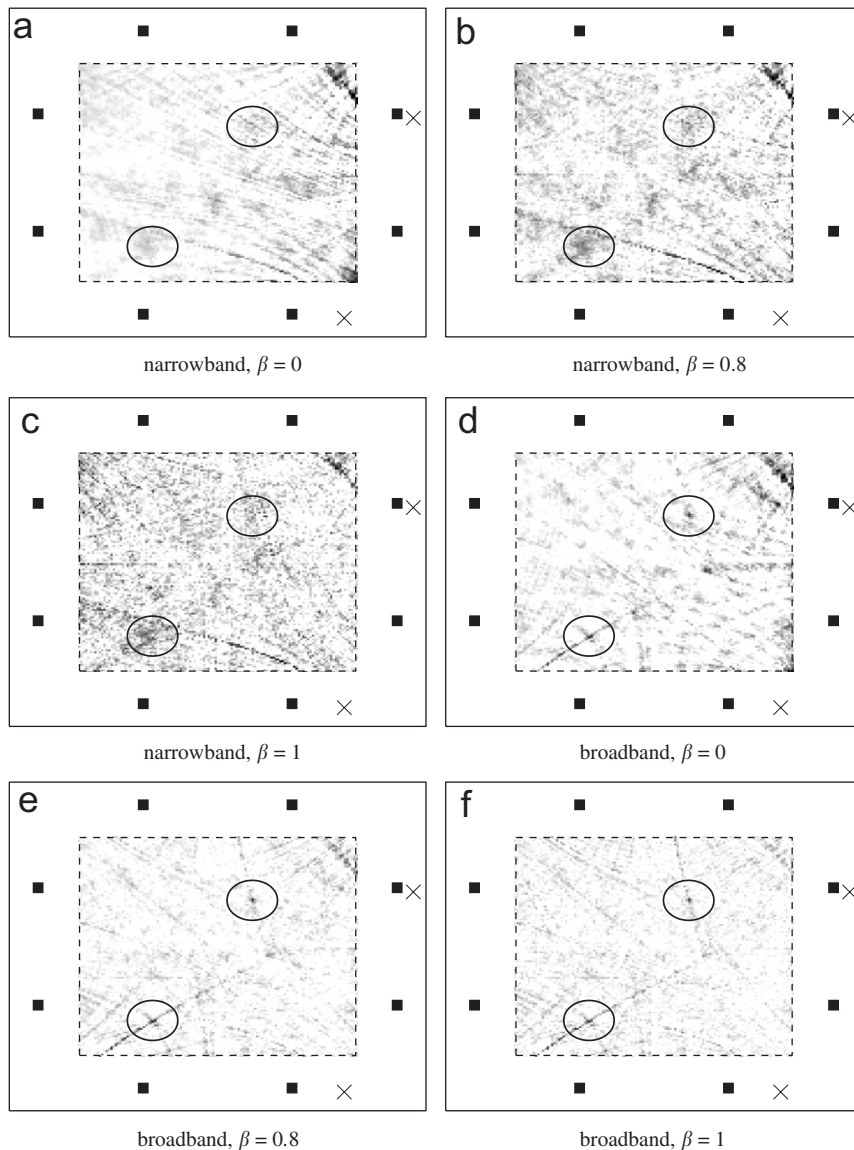


Fig. 2. Simulated SRP images using PHAT- β for two narrowband (a–c) and two broadband (d–f) target signals, positions denoted by circle centers. Noise sources are located at X markers and broken square denotes the FOV and square markers denote the microphone positions.

0.86. The effective SNR in this case was -12 dB. For each image the relative strengths of the target and noise peaks is observed for $\beta = 0, 0.8$, and 1.

Fig. 2(a) shows the target pixels in the circle center, and noise power distribution over the image for the narrowband target with $\beta = 0$. Noise peaks are observed from partial coherences at non-target positions. As β increases to 0.8 and 1 (Figs. 2(b) and (c), respectively) the partial coherences are reduced as the dominant noise peaks loose strength. In the narrowband signal case, however, the targets also

degrade from the PHAT. Increases in β increase the density of low-level fine-grained noise peaks along with the targets, while the amplitude of the large noise peaks (upper right corner) have reduced relative to the other peaks. Figs. 2(d), (e), and (f) show an analogous scenario for two broadband targets. In this case the target peaks appear to increase relative to the noise peaks for increasing β . The strong false peaks, as seen in Fig. 2(d), reduce relative to the highly coherent target peaks for increasing β . The spectral normalization from the

PHAT reduces the energy in the resonant modes from the partial coherences due to the reverberations.

Performance measure computations used the closest peak to the actual target position as target amplitude, provided that the pixels along the line connecting the peak to the original target position did not dip below 6 dB from the peak value and stayed positive. If the closest peak violated this condition, the target was considered undetected and its magnitude was set to zero. A noise peak was considered to be any peak outside a detected target area. This was the case if pixels on a line connecting the detected target peak to the potential noise peak included a non-positive pixel value or dropped 6 dB below the target peak. This ensured that perturbations along the gradients associated with the target peaks were not accepted as noise peaks.

For the A_z computations, a 1–8 ratio was maintained between the number of target and noise peaks, by only using the largest noise peaks. If all peaks were used, then many low-level peaks would be included resulting in a very low false positive

ratio, and thereby skewing the A_z statistic high by virtue of simply increasing the number of low-level noise peaks. To eliminate this dependency and to emphasize the detection relative to the worst-case noise peaks, the ratio between target and noise peaks was fixed.

Each plot in Figs. 3–8 was generated from 64 independent simulations of SRP images, which included random positions of target and noise sources, and random generation of noise signals. A 20 ms interval was used for Δ_I and the interval was placed 0.5 s from the start of the noise signals to ensure the steady-state room reverberations were included as well as early reflections of the target. Incoherent noise of -25 dB was added to each microphone signal (relative to the mean power in direct path signal) to simulate system/sensor noise. In the case of the real signals, shown in Figs. 7 and 8, the 20 ms interval was randomly shifted over a range of 12.5 ms within the phoneme for each Monte Carlo run.

The direct path signal to each microphone was computed separately from the noise signals so SNR

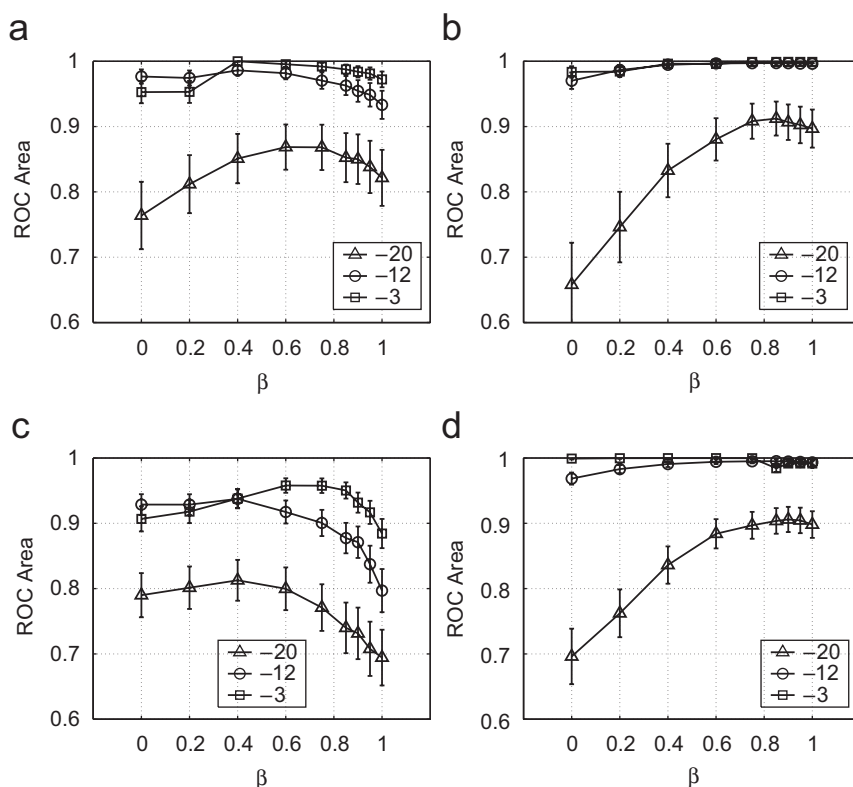


Fig. 3. ROC areas with 95% confidence limit bars from simulation target detection. The legend denotes SNR with room reflectivity set to 0. (a) Single narrowband target. (b) Single broadband target. (c) Two narrowband targets. (d) Two broadband targets.

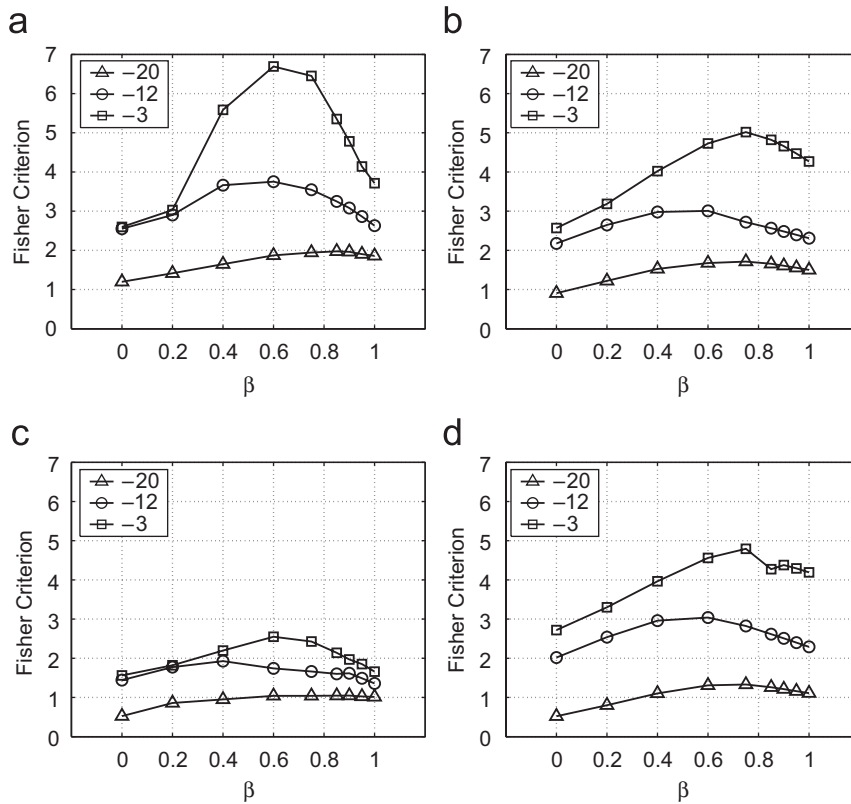


Fig. 4. FC from simulated target and noise images for 1 and 2 target cases. The legend denotes SNR with room reflectivity set to 0. (a) Single narrowband target. (b) Single broadband target. (c) Two narrowband targets. (d) Two broadband targets.

values (based on equation 18) could be set by scaling the direct path signal relative to the coherent noise and reverberant target energy. In the cases where SNR was held constant and the reflection coefficients were increased, a scaling factor was applied to the coherent noise sources so the combined target reverberation and coherent noise (with its reverberation) resulted in the desired SNR. Computations of array input SNR values were set based on $\beta = 0$ (i.e. no PHAT) and computed on a per target basis (SNR value applies to the average target strength in the case of multiple targets).

5. Simulation results

This section presents the simulation results in a series of plots. The relationship to β and performance for the different cases is highlighted and explained. The section concludes with generalizations for selecting β values for robust performance.

Fig. 3 shows A_z estimates for narrow and broadband targets, with no room reverberations.

The SNR is due only to coherent noise. The β value resulting in the best performance varies over the different cases shown. The most dramatic changes occur for the low SNR cases. More dramatic performance improvements for increasing β are observed for the broadband signal (almost 30% improvement for worst-case SNR), while very little performance differences are observed between the 1 and 2 target cases (within the 95% confidence limits). For the narrowband signals an approximate 5% performance loss is observed between the 1 and 2 target case with more dramatic performance losses as β approaches 1. For low SNR cases, best β values for the broadband targets occurred near 0.8, and for narrowband targets best β values ranged between 0.4 and 0.8.

Similar performance trends can be observed in Fig. 4 for the FC, with optimal performance occurring for β values in the 0.6–0.8 range. Both narrow and broadband cases exhibit performance loss for β approaching either 1 or 0, where these losses tend to be more dramatic for higher SNR

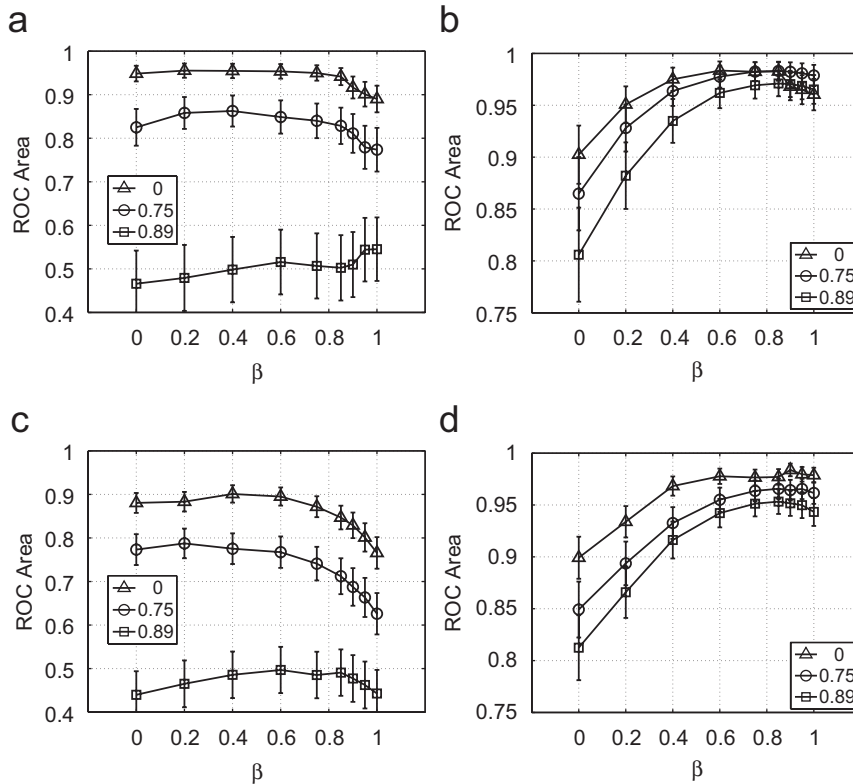


Fig. 5. ROC areas with 95% confidence limit bars from simulation target detection. The legend denotes the mean room reflectivity where the SNR was kept constant at -15 dB. (a) Single narrowband target. (b) Single broadband target. (c) Two narrowband targets. (d) Two broadband targets.

values. This is in contrast to the A_z cases, which saturate near 1 once the SNR exceeds a certain level. For both measures, however, a beta of 0 (no PHAT) or 1 (standard PHAT) does not result in the best performance for either signal (except for high SNR broadband signals where detection performance saturates at 1 over a range of β values).

The next set of results, shown in Figs. 5 and 6, fix the overall SNR to -15 dB (slightly below -12 dB case in previous figures) and include room reverberation. In order to maintain a fixed SNR with increasing reverberation noise, the coherent noise scatterer power was decreased. The correlation between the reverberation and the original target signal increases the chances of partial coherences in the SRP noise pixels. As a result, dramatic performance decreases can be observed for the narrowband targets in both Figs. 5 and 6 for a fixed SNR and increasing reflection coefficients. In addition, the A_z values for the narrowband signals of Fig. 5(a) show no significant enhancement from the PHAT- β . For the narrowband case A_z falls to

0.5 for the highest reverberation level, which indicates that the target and noise peak cannot be consistently distinguished from each other. The 2-target case in Fig. 5(c) shows more dramatic degradation as β approaches 1, while this trend is not observed for the FC. This indicates that the PHAT is not as effective at reducing the large amplitude partial coherent points (affecting the detection performance) as it is in overall reducing the noise peaks (affecting the mean and variance estimates of the FC). In contrast, the broadband signal results show dramatic improvement in the A_z values for increasing β , consistent with previous reports in the literature. For the narrowband cases the best β values tend to fall between 0.4 and 0.6, while for the broadband signals best performance is achieved for β values greater than 0.5.

Finally, Figs. 7 and 8 show results for a recorded voice signal segment. The SH sound is similar to the broadband signal examined in previous simulations. The spectrum for this sound is relatively flat (although not as flat as the simulated signal). Unlike

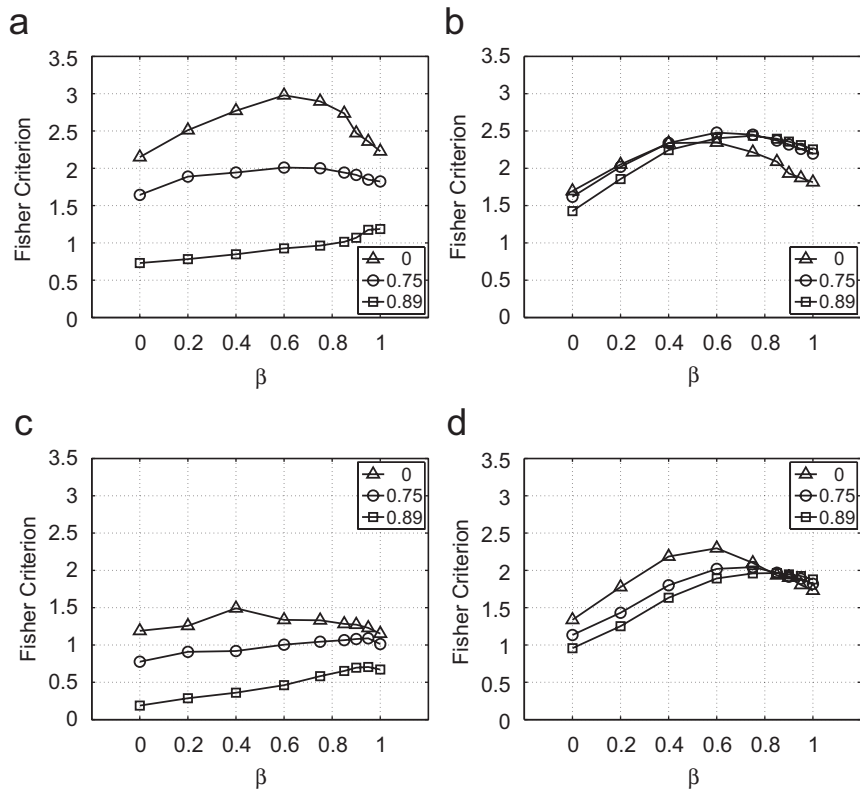


Fig. 6. FC from simulated target and noise images for 1 and 2 target cases. The legend denotes the mean room reflectivity where the SNR was kept constant at -15 dB. (a) Single narrowband target. (b) Single broadband target. (c) Two narrowband targets. (d) Two broadband targets.

the simulated signal, the SH is more noise like, with energy uniformly distributed over the whole time interval. The OU is more resonant like the narrowband signal. From 100 to 500 Hz the OU spectrum exhibits strong comb-like peaks that roll-off rapidly for increasing frequencies. So unlike the simulated narrowband signal, it is comb-like with limited, but non-zero, signal energy outside the main band, and is distributed over the whole time interval.

Figs. 7(a) and (c) show the A_z values for the OU sound with coherent and reverberation noises. These exhibit a similar performance to the single narrowband signal cases of Figs. 3 and 5. In particular, note the performance roll-off as β approaches 1 in all cases, where the roll-off is more significant for the real signal cases. Fig. 7(c) shows that the performance for the real signal does not change significantly at low β with respect to reverberation, while the simulated narrowband signal of Fig. 3(a) shows significant degradation. For high β values the real signal performance drops dramatically to that of the simulated cases. The

other plots show strong similarities between the real speech and simulated Butterworth signals. The main difference between the narrowband signal and the real speech at low β is that the real speech signal is distributed over time (quasi-periodic) as well as frequency. This provides a level of resistance to reverberation degradation.

The detection performance for the SH sound shown in Figs. 7(b) and (d) exhibit similar patterns as the broadband signals in Figs. 3 and 5. The performance for the real signal is slightly less than that of the simulated case, which is due to increased reverberation noise from strong signals in previous time intervals. In case of the real signal, the time interval used was around the middle of the phoneme and therefore also contained reverberation from previous parts of the signal. The best overall β values range from 0.5 (limited by low SNR OU sound of Fig. 7(b)) to 0.6 (limited by narrowband high reverberation of Fig. 7(c)). In a similar manner a comparison of the FC for the SH sound in Figs. 8(b) and (d) with that in Figs. 4(b) and 6(b)

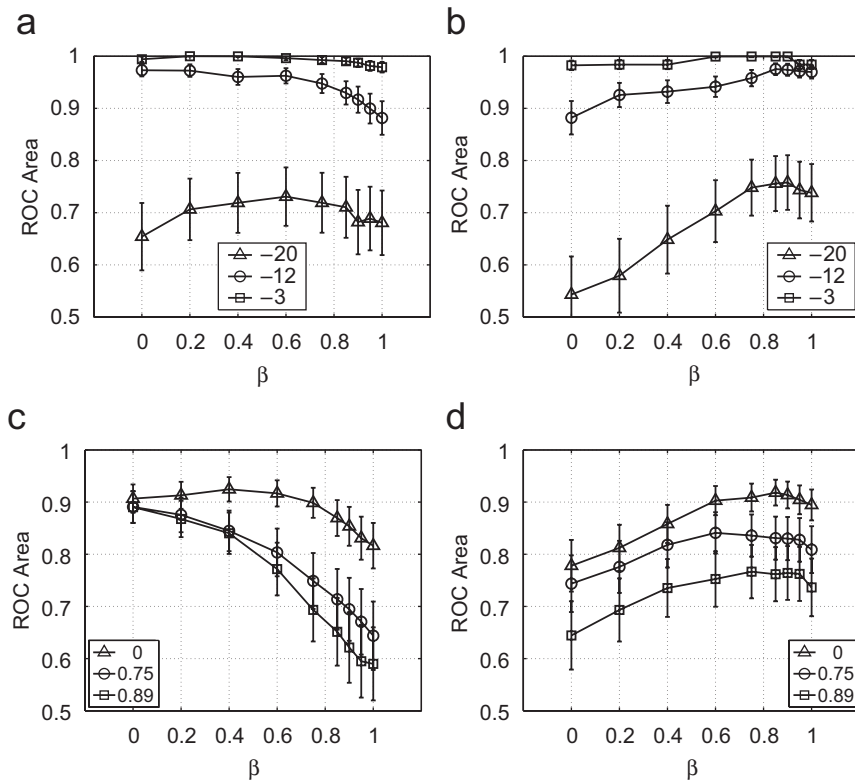


Fig. 7. ROC areas with 95% confidence limits for a real voice signal (“shoe”) simulated to have different positions and room conditions. (a) Single real target voiced speech OU. Legend represents SNR with no room reflections. (b) Single real target unvoiced speech SH. Legend represents SNR with no room reflections. (c) Single real target voiced speech OU. Legend represents mean reflectivity coefficient of -15 dB. Single real target unvoiced speech SH. Legend represents mean room reflectivity coefficient with an SNR of -15 dB.

shows a similar performance behavior in response to changes in β .

The simulation results highlight the ability of the PHAT to enhance detection performance for single or multiple targets in high levels of coherent and reverberant noise when the signal covers most of the spectral range. The simulated broadband signal covered approximately 65% of the system bandwidth, while the narrowband signal covered approximately 4% of the system bandwidth. Narrowband signals can sometimes benefit from the PHAT- β by using β values closer to 0.5. This enhances the target and noise amplitude separation (as shown in the FC). For β values greater than 0.5 degradations in detection performance typically occur, especially for multiple target in reverberant environments.

The results demonstrate that reverberation noise has a more detrimental impact on the detection and target–noise peak separation than coherent noise. The reverberant noise is correlated with the target

and enhances the target–noise term associated with Eq. (15). This increases the variance at the actual target position, and also for non-target positions by simply increasing the overall coherent noise power through additional multi-path energy. While significant performance improvements were observed for the PHAT applied to broadband signals in reverberant noise (Figs. 5(b), (d) and 7(d)), significant improvements were also observed for the low SNR cases of coherent noise only (Figs. 3(b), (d) and 7(b)). Therefore, the results indicate that PHAT improvements can be expected for coherent source noise degradations based on the uniform weighting of frequencies to improve the statistical averaging efficiency over the full spectrum.

Optimizations aimed at enhancing the SRP amplitude target position over the average noise amplitudes (i.e. related to FC) enhance the FC values, but not necessarily the detection performance as seen through the ROC analysis. As long as a frequency bin has sufficient SNR, it is of benefit to

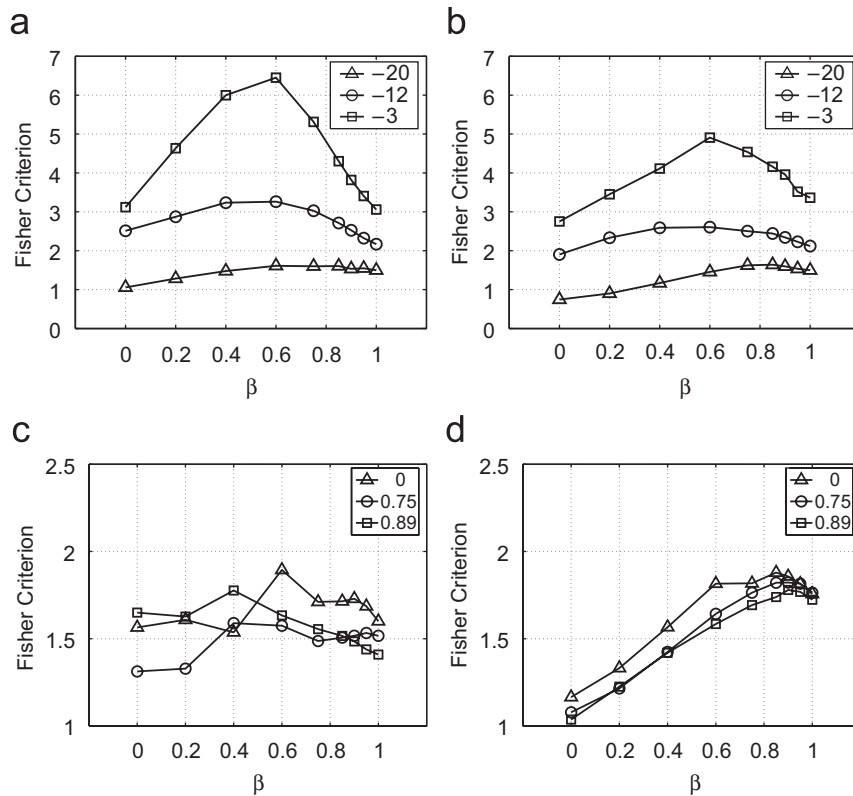


Fig. 8. The FC for a real voice signal (“shoe”) simulated to have different positions and room conditions. (a) Single real target voiced speech OU. Legend represents SNR with no room reflections. (b) Single real target unvoiced speech SH. Legend represents SNR with no room reflections. (c) Single real target voiced speech OU. Legend represents mean reflectivity coefficient of -15 dB. (d) Single real target unvoiced speech SH. Legend represents mean room reflectivity coefficient with an SNR of -15 dB.

scale it up to the same order of magnitude with other frequency bins and microphone channels to get the most efficient statistical averaging. The FC typically showed more β sensitivity than the ROC areas for single target and high SNR cases, especially seen in Figs. 4(a), (b), 8(a), and (b). Increasing the separation between noise and target peaks usually results in a better looking image as the noise values get smaller on average. At some point this increase no longer helps the detection performance relative to the dominant noise peaks as seen by the flattening of the ROC areas of these regions.

For low SNR cases, the FC shows insensitivity, while the ROC areas show significant changes (compare Figs. 3(c) and 4(c) in the multi-target case). Significant losses in ROC areas are observed for β approaching 1, while the FC values change very little. This implies that the target peaks are degraded along with the noise peaks, however the

(few) dominant noise peaks are not degrading as fast, which directly affect the ROC area computations (this pattern is observed in Figs. 2(a)–(c)). Therefore, the ROC analysis provides important information for performance prediction and parameter optimization associated with automatic target detection applications. Enhancement of the separation between signal and noise peaks (directly related to output SNR) may not necessarily lead to an improvement in detection rates. In addition, limiting the ROC analysis to a small number of worst-case noise peaks prevents performance improvements that result from simply increasing the number of low-level noise peaks.

For applications of the PHAT- β to general audio source signals, a β value of 0.5 is a good compromise if equal occurrences of narrow and broadband signals with similar SNR values can be expected. In cases similar to speech, the narrowband

voiced components typically are quasi-periodic over the analysis window and dominate a smaller portion of the spectrum with comb-like spectral amplitude related to the signals periodicity. This suggests selecting a β value less than 0.5. Figs. 7(a) and (c) indicate a slightly improved performance at this point when no reverberation is present. For increased reverberation, there is a clear performance loss (about 10%) from the no-PHAT or $\beta = 0$ case. However, for β greater than this there is a dramatic performance loss. So $\beta < 0.5$ avoids the dramatic performance degradations for the voiced speech, while a $\beta > 0.4$ maintains some enhancement for the unvoiced speech. To improve the detection rates over the whole signal in this case, greater preference should be given to weaker signals (typically the unvoiced speech components), and β values larger than 0.5 should be considered. Therefore, for speech a β value between 0.5 and 0.6 can result in significant enhancements for the weaker unvoiced signals while limiting the degradation for the stronger voiced signals at reasonable SNR levels.

6. Conclusions

The paper introduced an ROC area analysis for assessing the detection performance of SSL processes. Comparisons between noise and target separations based on FC showed different performance sensitivities than the ROC area statistics, suggesting the importance of optimizations based on the ROC analysis with fixed ratios of target and worst-case noise peaks. The simulation experiments for narrow and broadband signal sources showed a consistent detection performance loss when using the PHAT with narrowband signals. The losses were most significant for high reverberation and multi-target environments. A significant performance improvement was consistently shown for broadband targets especially in low SNR environments. The PHAT- β allowed for a parametric variation on the amount of influence given to the original spectral amplitude on the final coherent power values. This was useful for studying the impact of the PHAT and as a way to make the performance of the PHAT more robust over a range of narrow and broadband target sources. Simulation results suggested that for complex signal sequences with variable bandwidth, β values between 0.5 and 0.6 should be used for robust performance enhancement.

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